Multitask Assessment of Multilingual Email Writing - short description

Ritwik Ghosal

***Abstract*—This work develops a multilingual system for jointly predicting grammar quality, content coverage, and CEFR proficiency from an email writing prompt and its response. The five target languages are Dutch, Greek, Italian, Polish, and Portuguese. The training data combines a controlled synthetic corpus, designed around the supplied scoring rubrics, with authentic learner texts from public CEFR corpora. XLM-RoBERTa-base is adapted using LoRA and three task-specific prediction heads. Grammar and content are treated as bounded regression tasks, while CEFR is modelled as an ordered classification problem. Experiments compare first-token and attention-mask-aware mean pooling. The final interface supports either model independently and a default task-specific ensemble. Complete data construction, training, and evaluation details are provided in the accompanying technical report.**

I. Task and Design Motivation

The task requires three related but distinct judgements. Grammar measures linguistic correctness, content measures coverage of the requested information, and CEFR represents broader language proficiency. A response may therefore be grammatically strong but incomplete, or detailed but linguistically weak. For this reason, the three targets were learned jointly while retaining separate prediction heads.

XLM-RoBERTa-base was selected because it provides a shared multilingual encoder for all five languages. Parameter-efficient fine-tuning with LoRA reduced the number of trainable encoder parameters and made the experiments practical on a single NVIDIA T4 GPU.

II. Training Data

The final dataset contains 4,000 examples, divided into 3,200 training and 800 validation rows. It combines 2,000 controlled examples with 2,000 authentic learner texts.

TABLE I
Composition of the final training dataset.

Source	Rows
Controlled multilingual data	2,000
UniversalCEFR: ELG CEFR Dutch	645
UniversalCEFR: MERLIN Italian	600
UniversalCEFR: COPLE2 Portuguese	446
UniversalCEFR: PEAPL2 Portuguese	309
Total	4,000

The controlled corpus began with 40 practical email scenarios. Each scenario was explicitly represented in all five languages, producing 200 clean multilingual bases. An email was decomposed into functional components such as purpose, context, requested action, supporting information, and confirmation. These components were linked to atomic task details so that content could be degraded by removing requested information rather than by simply shortening the text.

Content scores from 0 to 4 were generated from the retained task details. Grammar variants were produced separately using typed language-specific transformations. These included agreement errors, incorrect articles or prepositions, verb-form errors, missing diacritics, fragments, and limited spelling noise. Each injected error was stored with its type, severity, original form, and replacement. Identifiers, names, dates, and reference codes were protected during corruption.

The authentic corpora added naturally occurring learner language and reduced dependence on synthetic surface patterns. Exact CEFR labels were used directly. Where a source provided only a band such as A1–A2, the label was retained as an interval rather than converted to an unjustifiably precise class. Grammar and content losses were masked for authentic rows without those annotations. When an original writing prompt was unavailable, the prompt input segment was left empty rather than fabricating task information.

Scenario-level separation was used for controlled data so that localized or corrupted versions of the same scenario could not cross between training and validation. Authentic texts were separated by source identity or text identity.

III. Model and Training

The prompt and email are tokenized as a paired sequence and processed by XLM-RoBERTa-base. LoRA adapters with rank 8 and scaling factor 16 are applied to the query and value projections. A shared projection maps the encoder representation to a 256-dimensional feature vector.

Three heads operate on this representation. Grammar and content use sigmoid outputs scaled to their rubric ranges, 0–5 and 0–4 respectively. CEFR uses an ordinal head with five learned, strictly ordered thresholds. The number of thresholds passed maps to A1, A2, B1, B2, C1, or C2.

The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{grammar}} + \mathcal{L}_{\text{content}} + \mathcal{L}_{\text{CEFR}}. \tag{1}$$

Grammar and content use normalized mean squared error. CEFR uses binary cross-entropy over the five ordinal threshold decisions. Interval labels supervise only decisions that are unambiguous for the complete interval. Missing task labels are excluded through masks.

Training used AdamW, mixed precision, gradient clipping, linear warmup and decay, and validation-based checkpoint

TABLE II
INTERNAL VALIDATION RESULTS FOR THE HYBRID MODELS.

Metric	First token	Mean pooling
Grammar Pearson	**0.9208**	0.9063
Content Pearson	0.9710	**0.9767**
CEFR soft Pearson	**0.8702**	0.8580
CEFR exact accuracy	0.5977	0.6256
Calibrated CEFR accuracy	0.6674	**0.6767**

selection. LoRA parameters used a learning rate of 2×10^{-4}, while the shared projection and prediction heads used 5×10^{-4}. Both groups used a weight decay of 0.01. Models were trained for eight epochs.

IV. Experiments and Validation Results

Four staged experiments were performed. The first two used only controlled data and compared five and eight training epochs. The later experiments used the hybrid dataset and compared first-token pooling with attention-mask-aware mean pooling.

The first-token model performed better for grammar, whereas mean pooling performed better for content and calibrated CEFR accuracy. This is plausible because individual grammatical cues can be captured through the encoder's learned global summary, while content details are distributed throughout the sequence.

CEFR cut-points were calibrated using only the internal validation partition. Calibration changed the conversion from the continuous ordinal score to a discrete class, but did not alter trained model parameters.

The default inference mode therefore uses the first-token checkpoint for grammar and the mean-pooling checkpoint for content and CEFR. Single-model first-token and mean-pooling modes are also provided for environments that prefer a smaller memory footprint.

V. Inference and Reproducibility

The repository contains model definitions, LoRA adapters, prediction heads, tokenizer files, configuration, prepared training data, clean training notebooks, and a standalone inference script. The interface accepts CSV or Excel files containing `questionStatement` and `text`, preserves row order, and writes all predictions to a new file.

Three modes are supported:

- `ensemble`: first-token grammar and mean-pooling content/CEFR;
- `cls`: first-token checkpoint for every output;
- `mean`: mean-pooling checkpoint for every output.

The complete repository archive was tested in a fresh Colab runtime. All three modes successfully restored their checkpoints, processed the provided example file, preserved row order, and produced predictions within the required ranges. Exact commands and file descriptions are provided in the README. The accompanying technical report contains the full data-generation procedures, error taxonomy, training design, calibration method, and extended analysis.

VI. Limitations and Conclusion

The controlled data provides precise supervision but cannot represent every naturally occurring learner error. Authentic data improves linguistic realism, although its language and proficiency distributions are uneven and it does not provide grammar and content labels. Consequently, internal validation results should not be interpreted as a complete estimate of performance under every domain shift.

The final system demonstrates a reproducible multilingual multitask pipeline that combines rubric-controlled supervision with authentic learner writing. Its main design choices are typed data generation, scenario-level leakage prevention, masked multitask learning, ordinal CEFR prediction, and lightweight LoRA adaptation.

a) Use of AI assistance: Large language model assistance was used for brainstorming, code scaffolding and debugging, multilingual drafting, and documentation editing. The author was responsible for the final design choices, implementation, execution, validation, and interpretation.

Please go through the text (detailed assessment) for the full descriptions.